LAB 1

#1 WAP to display “Hello World” on console display

```
#include <iostream>
using namespace std;
int main(){
        cout<<"Hello World";
return 0;
}
```

```
hello world

...Program finished with exit code 0
Press ENTER to exit console.
```

#2 WAP that reads two values and displays larger value

```
#include <iostream>
using namespace std;
int main() {
        int
    num1,num2;
    cin>>num1>>num2;
    //using ternary operator
    (num1>num2) ? (cout<<"Larger is
"<<num1) : ((num2=num1) ?
(cout<<"Both are equal") :
(cout<<"Larger is "<<num2));
        return 0;
}
```

```
2 2
Both are equal

4 5
Larger is 5

5 4
Larger is 5
```

#3 WAP to read three values a,b,c, and display x = a/b-c

```
#include <iostream>
using namespace std;
int main() {
        float a,b,c;
    cin>>a>>b>>c;
    float x;
        x = a/b-c;
    cout<<x;
        return 0;
}
```

```
1 1 1
0

...Program finished with exit code 0
Press ENTER to exit console.
```

#4 WAP to implement control characters (\n, \t, \a, \r)

```
#include <iostream>
using namespace std;

int main() {
    cout<<"showing how to use \n to add new line and \t for tab space";
        cout<<"\n ring a bell using \a"<<endl;
    cout<<"carriage return brings cursor to \r beginning";

        return 0;
}
```

```
showing how to use \n
 to add new line and     \t for tab space
 ring a bell using \a
 beginningeturn brings cursor to \r

...Program finished with exit code 0
Press ENTER to exit console.
```

#5 Implement namespace in a program

```
#include <iostream>
using namespace std;
namespace one{
        int value = 100;
}
namespace two{
        int value=200;
}
int main(){
    cout<<"from first namespace: "<<one::value<<endl;
    cout<<"from second namespace: "<<two::value<<endl;
        return 0;
}
```

```
from first namespace: 100
from second namespace: 200


...Program finished with exit code 0
Press ENTER to exit console.
```

#6 WAP to define a structure student that contains 3 variables and 2 functions

```
#include <iostream>
using namespace std;
struct student{
        string
    name; int
    roll;
        int marks;

void setstudent(){
        cout << "Enter name: ";
        cin >> name;
        cout << "Enter roll number: ";
        cin >> roll;
        cout << "Enter marks: ";
        cin >> marks;
}
void getstudent(){
        cout << "\nName: " << name;
        cout << "\nRoll Number: " <<
    roll; cout << "\nMarks: " << marks <<
    endl;
}
};
int main(){
        student s[3];
        for (int i = 0; i < 3; i++){
                cout << "\nEnter details of
        student "
<< i + 1 << endl;
                s[i].setstudent();
        }
        for (int i = 0; i < 3; i++){
                cout << "\nDetails of student
        " << i
+ 1 << endl;
```

```
Enter details of student 1
Enter name: gauri
Enter roll number: 211
Enter marks: 94

Enter details of student 2
Enter name: aadhya
Enter roll number: 212
Enter marks: 95

Enter details of student 3
Enter name: mansha
Enter roll number: 213
Enter marks: 96

Details of student 1

Name: gauri
Roll Number: 211
Marks: 94

Details of student 2

Name: aadhya
Roll Number: 212
Marks: 95

Details of student 3

Name: mansha
Roll Number: 213
Marks: 96
```

#7 In the previous program make the data private and the functions public

```
#include <iostream>
using namespace std;
class student{
private:
        char
    name[50]; int
    roll; float
    marks;
public:
        void setstudent(){
                cout << "Enter name: ";
                cin >> name;
                cout << "Enter roll number:
        ";
                cin >> roll;
                cout << "Enter marks: ";
                cin >> marks;
        }
        void getstudent(){
                cout << "\nName: " << name;
                cout << "\nRoll Number: " <<
        roll; cout << "\nMarks: " << marks
        <<
endl;
    }
};
int main()
{
        student s[3];
        for (int i = 0; i < 3; i++){
        cout << "\nEnter details of student
" << i + 1 << endl;
                s[i].setstudent();
        }
        for (int i = 0; i < 3; i++){
                cout << "\nDetails of
```

```
Enter details of student 1
Enter name: gauri
Enter roll number: 1
Enter marks: 7

Enter details of student 2
Enter name: aadhya
Enter roll number: 2
Enter marks: 8

Enter details of student 3
Enter name: mansha
Enter roll number: 3
Enter marks: 9

Details of student 1

Name: gauri
Roll Number: 1
Marks: 7

Details of student 2

Name: aadhya
Roll Number: 2
Marks: 8

Details of student 3

Name: mansha
Roll Number: 3
Marks: 9
```

#8 Convert program #6 from struct to class

```
#include <iostream>
using namespace std;

class student {
private:
        string
    name; int
    roll; int
    marks;

public:
        void setstudent() {
                cout << "Enter name: ";
                cin >> name;
                cout << "Enter roll number:
        ";
                cin >> roll;
                cout << "Enter marks: ";
                cin >> marks;
        }

        void getstudent() {
                cout << "\nName: " << name;
                cout << "\nRoll Number: " <<
        roll; cout << "\nMarks: " << marks
        <<
endl;
    }
};

int main() {
        student s[3];

        for (int i = 0; i < 3; i++) {
        cout << "\nEnter details of student
" << i + 1 << endl;
```

```
Enter details of student 1
Enter name: g
Enter roll number: 1
Enter marks: 4

Enter details of student 2
Enter name: a
Enter roll number: 2
Enter marks: 5

Enter details of student 3
Enter name: m
Enter roll number: 3
Enter marks: 7

Details of student 1

Name: g
Roll Number: 1
Marks: 4

Details of student 2

Name: a
Roll Number: 2
Marks: 5

Details of student 3

Name: m
Roll Number: 3
Marks: 7
```

```
+ 1 << endl;
                s[i].getstudent();
        }

        return 0;
}
```

#9 Discuss the differences of structures in C and structures in C++

In C structures are simple data structures that are only used to hold and display data items, they don't contain functions. But in C++, structures contain functions.

Also in C++ the structure is treated as a class, we can have public/private members.

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#10 WAP to read integer value and display on screen "WELL DONE" that many times.

```
#include <iostream>
using namespace std;

int main() {
        int x;
    cout<<"Enter how many times to
print: ";
        cin>>x;
            for(int i=0; i<x; i++)
        cout<<"WELL DONE"<<endl;
        return 0;
}
```

```
Enter how many times to print: 3
WELL DONE
WELL DONE
WELL DONE


...Program finished with exit code 0
Press ENTER to exit console.
```